

ECONOMETRICS I

Problem Set 3

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1. **Argue extensively** whether each of the following statements is true, false, or uncertain based on technical arguments, graphs, and/or formulas.
 - (a) Regressions with excess regressors tend to be inefficient. However, the exclusion of certain variables could also lead to high variance.
 - (b) In a univariate OLS regression satisfying all necessary conditions, the more the regressor varies in repeating sampling, the better (more efficient) the estimation will be.
 - (c) If we obtain a coefficient not significantly different from zero in a regression, we should conclude that this regressor is not relevant in the data generating process.
 - (d) The p-value of a coefficient is a better measure of precision than the standard deviation of a coefficient.
 - (e) Homoskedasticity in the data generating process is an essential assumption ensuring the unbiasedness of the OLS estimator.
 - (f) When estimating a regression that includes only irrelevant or non-causal variables, it is impossible to obtain significant coefficients.
 - (g) When assuming $\mathbb{E}[u|X] = 0$, one implicitly assumes that all variables excluded from a regression (relevant or irrelevant) do not correlate with the variables included in it.
 - (h) Vertical and horizontal outliers are influential in the slopes of an OLS estimation in different ways because the estimator minimizes vertical residuals and not horizontal ones.

2. For each concept in the left column, indicate all related concepts from the right column.

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| (i) Low number of observations | (A) Unreliable coefficients |
| (ii) High multicollinearity | (B) Inefficiency |
| (iii) Omission of relevant variable not related to regressors | (C) High variance of the error term |
| (iv) Omission of relevant variable related to regressors | (D) Bias |
| (v) Inclusion of irrelevant variable | (E) Invalidity of the coefficient variance-covariance matrix |
| (vi) Heteroscedasticity | (F) Invalidity of a projection variance-covariance matrix |
| (vii) 'Horizontal' outliers | |
| (viii) Simultaneous causality | (G) Invalid confidence interval |
| (ix) Non-normality of errors | (H) Coefficient vector not defined |